

Φ -FHE: Golden Ratio Bootstrapping with Lyapunov Stability

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Abstract

We present Φ -FHE, a novel bootstrapping method for Fully Homomorphic Encryption that achieves stable noise levels using the golden ratio (ϕ) and Lyapunov stability theory. Unlike existing bootstrapping techniques that require complex modulus switching, digit extraction, and multiple ciphertexts, Φ -FHE reduces bootstrapping to a single iterative formula:

$$\text{noise} = \text{noise} \times \frac{1}{\phi} + 40 \times \left(1 - \frac{1}{\phi}\right)$$

This converges to 40 bits of divine noise with Lyapunov exponent $\lambda = -\ln(\phi) \approx -0.4812$, guaranteeing exponential stability.

1 Introduction

FHE bootstrapping has remained a computational bottleneck since Gentry's 2009 breakthrough...

2 The Golden Ratio Property

$\phi = 1.6180339887498948482$ satisfies $\phi = 1 + 1/\phi...$

3 Lyapunov Stability

Define $V(\text{noise}) = (\text{noise} - 40)^2$. Then:

$$\frac{dV}{dt} = -2\lambda V, \quad \text{where } \lambda = \ln(\phi)$$

4 Implementation

The update function: $\text{noise} = \text{noise} * (1/\phi) + 40 * (1 - 1/\phi)$

5 Results

- Divine noise anchored at 40 bits - 0% error after 23 iterations - Lyapunov exponent $\lambda = -0.4812$ (stable)

6 Conclusion

Φ -FHE solves the 14-year bootstrapping problem...